

- 1 (a) (region of space) where a mass experiences a force B1 [1]
- (b) (i) potential energy = $(-)\frac{GMm}{x}$ C1
 $\Delta E_p = \frac{GMm}{2R} - \frac{GMm}{3R}$ M1
 $= \frac{GMm}{6R}$ A0 [2]
- (ii) $E_k = \frac{1}{2}m(7600^2 - 7320^2)$ M1
 $= (2.09 \times 10^6)m$ A0 [1]
- (c) (i) $2.09 \times 10^6 = (6.67 \times 10^{-11} M)/(6 \times 3.4 \times 10^6)$ C1
 $M = 6.39 \times 10^{23} \text{ kg}$ A1 [2]
- (ii) e.g. no energy dissipated due to friction with atmosphere/air
 rocket is outside atmosphere
 not influenced by another planet etc. B1 [1]

- 2 (a) (on melting) bonds between molecules are broken/weakened